

Smart Waste Classification Using AI

Project Overview

Smart Waste Classification Using AI is a machine learning project that automatically classifies waste into different categories using image recognition. The system helps improve waste segregation, supports recycling, and reduces manual effort by identifying the type of waste from an uploaded image.

Features

- Automatic waste image classification
- AI-based prediction
- Supports multiple waste categories
- Easy to use and scalable
- Helps in proper waste management

Technologies Used

- Python
- TensorFlow / Keras (or YOLOv8 if used)
- OpenCV
- NumPy
- Google Colab / VS Code

Dataset

The model is trained on a waste image dataset containing different categories such as:

- Plastic
- Paper
- Glass
- Metal
- Cardboard
- Organic Waste

Project Structure

- dataset/ – Training and testing images
- train.py – Model training script
- predict.py – Image prediction script
- model/ – Trained AI model
- README.md – Project documentation

Applications

- Smart Dustbins
- Recycling Centers
- Municipal Waste Management
- Educational AI Projects

Future Enhancements

- Real-time waste detection using a webcam
- Mobile application integration
- IoT-enabled smart waste bins
- Cloud-based monitoring system

Conclusion

This project demonstrates how Artificial Intelligence can be used to automate waste classification. It improves waste segregation efficiency, supports recycling initiatives, and contributes to a cleaner and more sustainable environment.



